

A NOTE ON CONCURRENT MONADS

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This note is taken from two papers on concurrent monads and concurrent categories:

[Rivas and Gaskeloff (2019)

Monads with merging]

[Rivas and Uustalu (2024)

concurrent monads for shared state]

Monads capture effects in programs. Concurrent monads have additional structure to run effects in parallel while respecting the "degree of sequentiality" of computations. This is achieved with an order structure that is compatible with the monoidal structure.

ORDERED MONAD

An order for a Set-monad (T, μ, η) is an assignment of a partial order $(TA, \leq_A)$ for each set A such that KlT is order-enriched with the partial order defined pointwise:

$$f \leq_{AB} g \text{ iff } \forall a \in A \quad f(a) \leq_B g(a) \\ \text{for } f, g: A \rightarrow TB.$$

ORDERED MONOIDAL FUNCTOR

An order for a lax-monoidal Set-endofunctor (T, m, e) is an assignment of a partial order $(TA, \leq_A)$ for each set A that is compatible with the monoidal structure:

for all $a, a' \in TA$ and all $b, b' \in TB$,
if $a \leq_A a'$ and $b \leq_B b'$,
then $m_{AB}(a, b) \leq_{A \times B} m_{AB}(a', b')$.

CONCURRENT MONAD

A concurrent monad is an ordered monad $(T, \mu, \eta, \leq)$ whose underlying functor is an ordered monoidal functor $(T, m, \eta_1, \leq)$ such that

$$(f \otimes f'); (g \otimes g') \leq (f; g) \otimes (f'; g') \\ \text{for all } f: A \rightarrow B, f': A' \rightarrow B', \\ g: B \rightarrow C \text{ and } g': B' \rightarrow C' \text{ in } KlT.$$

CONCURRENT CATEGORY

A concurrent category is an identity-on-objects poset-enriched functor $J: \mathcal{P} \rightarrow \mathcal{C}$ from a poset-enriched monoidal category, $(\mathcal{P}, \otimes, I, \leq)$, to a poset-enriched category, $(\mathcal{C}, \leq)$, with a morphism $\text{id} \in \mathcal{C}(I, I)$ and an operation $\parallel: \mathcal{C}(A, B) \times \mathcal{C}(A', B') \rightarrow \mathcal{C}(A \otimes A', B \otimes B')$ satisfying

- $\text{id}_I \leq \text{id}$
- $\text{id}; \text{id} \leq \text{id}$
- $\text{id}_{x \otimes y} \leq \text{id}_x \parallel \text{id}_y$
- $(f \parallel f'); (g \parallel g') \leq (f; g) \parallel (f'; g')$
- $J(f \otimes f'); (g \parallel g') = (Jf; g) \parallel (Jf'; g')$
- $(f \parallel f'); J(g \otimes g') = (f; Jg) \parallel (f'; Jg')$

CONCURRENT MONAD

A concurrent monad, (T, μ, η, m, e) , on a poset-enriched monoidal category, $(\mathcal{P}, \otimes, I, \leq)$, is a monad, (T, μ, η) , whose underlying functor is lax-monoidal, (T, m, e) , and poset-enriched; moreover, the multiplication and unit must be laxly monoidal:

- $e \geq \eta_I$
- $e \geq e; Te; \mu_I$
- $(\eta_x \otimes \eta_y); m_{x,y} \geq \eta_{x \otimes y}$
- $(\mu_x \otimes \mu_y); m_{x,y} \geq m_{x,y}; Tm_{x,y}; \mu_{x \otimes y}$

A concurrent monad is normal if $e = \eta_I$.

THEOREM

The Kleisli construction for a concurrent monad, T , on a poset-enriched monoidal category, $\mathcal{P}$, defines a concurrent category, $J: \mathcal{P} \rightarrow \text{Kl}T$.